

1 Cross Norms

Notation 1.1. In this section, A_1, A_2 will denote Banach spaces. The norms on these spaces will be denoted by $\|\cdot\|$ unambiguously as we will distinguish between elements of A_1 and A_2 .

Notation 1.2. For Banach spaces A_1, A_2 , we will denote their algebraic tensor product by $A_1 \otimes A_2$. Its completion in some norm $\|\cdot\|_\beta$ will be denoted by $A_1 \otimes_\beta A_2$.

Definition 1.3. A norm $\|\cdot\|_c$ on $A_1 \otimes A_2$ is called a cross norm if it satisfies $\|x_1 \otimes x_2\|_c = \|x_1\| \|x_2\|$ for all $x_1 \in A_1, x_2 \in A_2$.

Remark 1.4. That $\|\cdot\|_\lambda$ is a cross norm is immediate from an application of the Hahn Banach Theorem.

Suppose

Definition 1.5. The injective cross norm $\|\cdot\|_\lambda$ on $A_1 \otimes A_2$ is defined by

$$\|\xi\| = \sup\left\{\left|\sum_{k=1}^n f(x_{1,k})g(x_{2,k})\right| : \begin{array}{l} \xi = \sum_{k=1}^n x_{1,k} \otimes x_{2,k} = \xi, \\ f \in A_1^*, \|f\| = 1, \\ g \in A_2^*, \|g\| = 1 \end{array}\right\}.$$

Remark 1.6. Let $\|\cdot\|_\beta$ be a cross norm on $A_1 \otimes A_2$. Retaining the notation of the previous definition,

$$|f(x_1)| |g(x_2)| \leq \|x_1\| \|x_2\| = \|x_1 \otimes x_2\|_\beta.$$

So, the injective cross norm is the smallest cross norm on $A_1 \otimes A_2$.

Definition 1.7. The projective cross norm on $A_1 \otimes A_2$ is denoted by $\|\cdot\|_\gamma$ and defined as

$$\|\xi\| = \inf\left\{\sum_{k=1}^n \|x_{1,k}\| \|x_{2,k}\| : \xi = \sum_{k=1}^n x_{1,k} \otimes x_{2,k}\right\}.$$

Remark 1.8. Let $\|\cdot\|_\beta$ be a cross norm on $A_1 \otimes A_2$. For ξ in $A_1 \otimes A_2$, and $\epsilon > 0$, there exists a representation $\xi = \sum_{k=1}^n x_{1,k} \otimes x_{2,k}$ in $A_1 \otimes A_2$, such that

$$\|\xi\|_\beta \leq \sum_{k=1}^n \|x_{1,k} \otimes x_{2,k}\|_\beta = \sum_{k=1}^n \|x_{1,k}\| \|x_{2,k}\| \leq \|\xi\|_\gamma + \epsilon.$$

So, the projective norm majorises all cross norms. That it is itself a cross norm is evident from the computation

$$\|x_1\| \|x_2\| = \|x_1 \otimes x_2\|_\gamma \leq \|x_1 \otimes x_2\|_\lambda \leq \|x_1\| \|x_2\|.$$

The same computation reveals that all norms ‘between’ the two norms are cross norms. projective norm

1.1 C^* cross-norms

In general, the projective cross-norm is not a C^* norm [1, Remark 4.4]. So we consider

Definition 1.9. *The maximal C^* norm defined by*

$$\|x\|_{\max} = \sup\{\|\pi(x)\| : \pi \text{ is a representation of } A_1 \otimes A_2\}$$

The max norm is a cross norm.

Remark 1.10. *For $\xi = \sum_{k=1}^n x_{1,k} \otimes x_{2,k}$, in $A_1 \otimes A_2$, we have*

$$\|\pi(\xi)\| = \left\| \sum_{k=1}^n \pi(x_{1,k} \otimes x_{2,k}) \right\| \leq \sum_{k=1}^n \|\pi(x_{1,k} \otimes x_{2,k})\| \leq \sum_{k=1}^n \|x_{1,k} \otimes x_{2,k}\|.$$

This shows that $\|x\|_{\max} \leq \|x\|_{\gamma}$.

In addition, for any representation π_1 of A_1 and π_2 of A_2 , we have

$$\begin{aligned} \|\pi_1(x_1)\| \|\pi_2(x_2)\| &= \|\pi_1(x_1)\pi_2(x_2)\| = \|(\pi_1 \otimes \pi_2)(x_1 \otimes x_2)\| \\ &\leq \|x_1 \otimes x_2\|_{\max} \leq \|x_1 \otimes x_2\|_{\gamma} \leq \|x_1\| \|x_2\|. \end{aligned}$$

By choosing faithful representations π_1 and π_2 , the left hand side and right hand side become equal.

Definition 1.11. *The minimal C^* cross-norm $\|\cdot\|_{\min}$ is defined by*

$$\|\xi\|_{\min} = \sup\{\|(\pi_1 \otimes \pi_2)(\xi)\| : \begin{array}{l} \pi_1 \text{ is a representation of } A_1, \\ \pi_2 \text{ is a representation of } A_2 \end{array} \}.$$

It is immediate that $\|\cdot\|_{\min} \leq \|\cdot\|_{\max} \leq \|\cdot\|_{\gamma}$.

Remark 1.12. *To see that $\|\cdot\|_{\min}$ is a cross norm, we only need to show that $\|\cdot\|_{\Lambda} \leq \|\cdot\|_{\min}$. For f in A_1^* , there exist a representation π_1 of A_1 on a Hilbert Space $\mathcal{H}_1$ and ξ_1, η_1 in $\mathcal{H}_1$ satisfying $\|f\| = \|\xi_1\| \|\eta_1\|$ and $f(a) = \langle \eta_1, \pi_1(a)\xi_1 \rangle$ for all a in A_1 . Similarly, for g in A_2^* , there exist a representation π_2 of A_2 on a Hilbert space $\mathcal{H}_2$ and ξ_2, η_2 in $\mathcal{H}_2$ such that $g(b) = \langle \eta_2, \pi_2(b)\xi_2 \rangle$ for all b in A_2 .*

and $\|g\| = \|\xi_2\| \|\eta_2\|$. This means for $x = \sum_{k=1}^n x_{1,k} \otimes x_{2,k}$ in $A_1 \otimes A_2$, we have

$$\begin{aligned}
& |(f \otimes g)x| \\
&= \left| \sum_{n=1}^k f(x_{1,k})g(x_{2,k}) \right| \\
&\leq \left| \sum_{n=1}^k f(x_{1,k})g(x_{2,k}) \right| \\
&\leq \left| \sum_{k=1}^n \langle \eta_1, \pi_1(x_{1,k})\xi_1 \rangle \langle \xi, \pi_2(x_{2,k})\xi_2 \rangle \right| \\
&= \left| \sum_{k=1}^n \langle \eta_1, \pi_1(x_{1,k})\xi_1 \rangle \langle \xi, \pi_2(x_{2,k})\xi_2 \rangle \right| \\
&= \left| \sum_{n=1}^k \langle \eta_1 \otimes \eta_2, (\pi_1 \otimes \pi_2)(x_{1,k} \otimes x_{2,k})(\xi_1 \otimes \xi_2) \rangle \right| \\
&= |\langle \eta_1 \otimes \eta_2, (\pi_1 \otimes \pi_2)(x)(\xi_1 \otimes \xi_2) \rangle| \\
&\leq \|\eta_1\| \|\eta_2\| \|(\pi_1 \otimes \pi_2)(x)\| \|\xi_1\| \|\xi_2\| \\
&= \|f\| \|g\| \|x\|_{\min}
\end{aligned}$$

In other words, $\|x\|_{\gamma} \leq \|x\|_{\min}$.

Thus, $\|\cdot\|_{\min}$ is a C^* norm.

tensor products of ucp maps functoriality and universal property

References

- [1] Gilles Pisier. *Tensor Products of C^* -Algebras and Operator Spaces: The Connes–Kirchberg Problem*. London Mathematical Society Student Texts. Cambridge University Press, 2020.